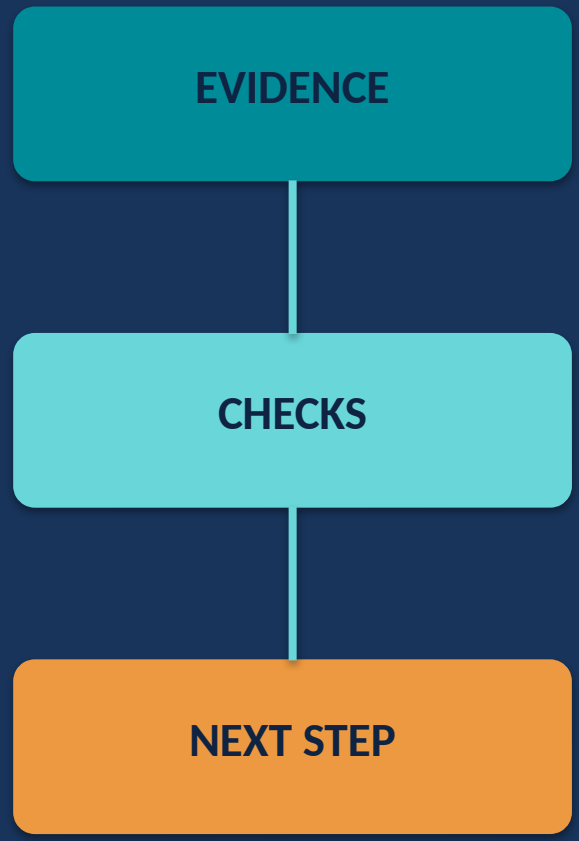
IBM Z DATATHON 2026 / AI SECURED

REFUND RESCUE

AI-Powered Refund Intelligence & Resolution System

Make refund status traceable. Keep evidence honest.

Challenge: AI Secured — Innovation Without Exposure



The problem: “Completed” is not “credited”

Customers need evidence of what happened—not another generic status message.

Merchant status

“Refund completed”

A reported status, not bank-credit proof.

Missing context

Reference IDs, timestamps, receipts and bank evidence may be scattered across channels.

Customer impact

Uncertainty, repeated support requests and time-consuming manual investigation.

The gap: one place to compare claims, evidence and verified data.

Refund Rescue helps organize and interpret evidence; it does not move money or guarantee a credit.

The solution: a Refund Proof Chain

A single case view that keeps each source clearly labeled.

1 CUSTOMER EVIDENCE

Receipts

Relevant statement excerpts

Customer notes

2 MERCHANT-REPORTED

Order/refund status

Refund date and reference

Merchant messages

3 VERIFIED SOURCE

Authorized provider/bank data

Confirmation timestamp

Only when integration exists

CORE RULE: A “refund completed” label is not proof of bank credit.

In the current demo, verified bank/provider data is not connected; examples use synthetic evidence.

From scattered evidence to a clear next step

Four practical capabilities designed around the customer’s real question: what should I do now?

01 Centralized case timeline

Bring order details, refund events, notes and uploaded evidence into one view.

02 Evidence-gap checks

Flag missing references, conflicting statuses and credit that remains unverified.

03 Guided next action

Suggest checking the processing window, contacting the merchant or escalating with evidence.

04 Complaint draft

Create a factual message for the customer to review, edit and send.

Current prototype: local browser demo • synthetic cases • transparent rule-based checks

Example case: what the system flags

Illustrative synthetic case—not a real customer transaction or a measured test result.

CASE RR-1042 • SYNTHETIC

Order amount

₹2,499.00

Merchant status

Refund completed

Refund reference

Not provided

Verified bank credit

Not connected / unverified

DETECTED GAPS

01 **Missing refund reference**

02 **No verified credit evidence**

RECOMMENDED NEXT STEP

Request the refund reference from the merchant; compare it with provider/bank records when available.

The rule flags uncertainty; it does not claim that the bank failed to pay.

Privacy by design: minimize exposure

Protect sensitive information before any future AI-assisted document analysis.

Controls to build and demonstrate

- Mask account numbers and unnecessary personal details.
- Use least-privilege, role-based access to cases.
- Keep customer evidence, merchant claims and verified records separate.
- Audit access and key case changes.

AI and data-handling guardrails

- Send only the minimum necessary masked content to an AI service.
- Do not use customer data for model training without appropriate authorization.
- Require user review before a complaint draft is sent.
- Use synthetic cases for demos and evaluation.

Current status: privacy principles are part of the design; production access controls and AI integration remain planned work.

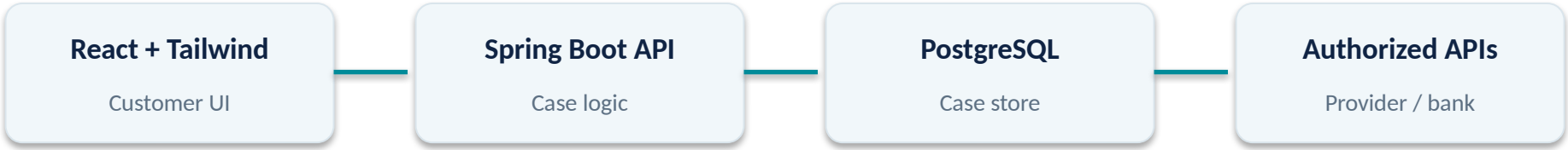
Architecture: prototype now, integrations later

Keep the current build separate from the proposed production architecture.

CURRENT DEMO

Browser UI (HTML/CSS/JavaScript) → Synthetic case data → Transparent rule checks

PROPOSED PATH



Optional masked LLM analysis • audit logs • access controls • integration only with permission

No live payment/bank API, backend database, live LLM, or LinuxONE deployment is claimed in this prototype.

Where IBM Z / LinuxONE fits

A potential deployment platform for a future secure, resilient service—not a completed deployment claim.

Secure processing

Host case-processing services in an enterprise environment with strong identity and access controls.

Operational resilience

Plan for dependable service, monitoring, auditability and recovery as usage grows.

Controlled integrations

Connect payment-provider or bank systems only through authorized, configured interfaces.

Honest scope: the demo runs locally; LinuxONE is a proposed future option.

Describe the security controls your team actually implements separately from capabilities of the target platform.

Evaluation plan: report evidence, not promises

Use labeled synthetic cases and publish measured results only after running the tests.

Detection precision

Of all flags raised, how many were correct?

Detection recall

Of all known gaps, how many were found?

Next-step time

Seconds needed to identify the recommended action.

Draft quality

Checklist score: factual, clear, relevant, no excess sensitive data.

Test categories: complete refund • missing reference • conflicting status • delayed/unverified credit

No accuracy, time-saved or resolution-rate numbers are claimed until the team executes and records the evaluation.

Expected value

A practical assistant for evidence organization and better-informed escalation.

For customers

Understand what is known, what is missing and what to ask next.

For support teams

Receive more structured cases with fewer repeated clarification requests.

For institutions

Get a clearer evidence trail when a case needs further investigation.

Value proposition: less guesswork, clearer evidence, more actionable next steps.

Refund Rescue supports resolution; it does not transfer funds, verify a bank account on its own, or guarantee a refund.

REFUND RESCUE

Make every refund traceable.

Privacy-aware refund tracking • Evidence-gap detection • Guided escalation

Challenge: AI Secured — Innovation Without Exposure

Team: Shooting Stars Foundation